

FIT3143 Applied #2 Assessment (Week 10)

GPU/NPU and Exponential Growth

OBJECTIVES

- Compare the fundamental principles of parallel hardware using vector processing, Graphics Processing Units (GPU) and Neural Processing Units (NPU); (ULO6)
- Judge modern parallel computing applications and the impact of exponential growth in hardware (ULO7)

MARKS

- The applied session is worth 10 of the unit's final mark.

APPLIED INSTRUCTIONS

1. Preparation is required for this Applied session. Please **DO NOT** plan to complete the session without any preparation/understanding.
2. Students should start working on the tasks at least 1 week prior to the assessed session.
3. **All students** must submit their answers and associated codes via Moodle in the required format before the start of the allocated class. A **late penalty** (5% per day) will apply if you do not submit the required materials on time! Students in the same team/group should submit the same set of files.
4. In addition to regular offline/off-class marking, students' (or Team's) slides, answers, and/or source code will also be reviewed and assessed through in-class presentations, peer reviews, and/or question-and-answer sessions.
5. For **team/group presentations**, marks in general will be allocated for:
 - The correctness of the design, codes, or results
 - The quality and clarity of the presentation
 - The correctness of the explanation during the presentation

The presentation content should include slides that are clear, easy to understand, and readable in a professional working environment (aka an industry setting). While the teaching team does not impose strict limits on the presentation format (slides/document format), nevertheless, if the presentation is hard to read/digest (e.g., speech is

misaligned with the presented content, or slides/speech are unclear, or arguments are made without sound logic), marks will be deducted. Both the quality and correctness of your submitted/demonstrated work are assessed and graded.

6. For **individual/group interviews and Q&A sessions**, marks will be allocated for:

- The correctness of the verbal answer
- The quality of the verbal answer

Focus on the most important idea in the answer. Marks will be awarded on the student's ability to explain their ideas with reference to the submitted code or documents in a concise, focused and accurate manner. Answers with errors, inaccurate or lengthy explanations will lose marks. Showing little to no understanding of the code will result in a loss of marks. Showing little to no understanding of the associated libraries, or submitting results will result in a loss of marks. Showing little to no understanding of how to select and filter information (aka dumping unfiltered answers without processing from the textbook/internet/AI agent) will lose marks.

- 7. Always make sure you read and follow the marking rubric well in advance of the deadline.
- 8. Marks will not be awarded if you skip the class, or do not make any submissions, or do not make any contributions in the working group/team, or make an empty submission to Moodle (unless a special consideration extension has been approved).
- 9. You are allowed to use Generative-AI to search for information and resources during the preparation period. However, you must declare in your report and upload all the prompt records (in PDF files).
- 10. AI tools are not allowed during the presentation period or any oral/coding interview sessions.
- 11. All submitted files should ideally include students' names, ID and Monash email addresses.

APPLIED IN-CLASS ASSESSMENT ACTIVITIES

Students are required to form teams of 2.

Note: For students who cannot form teams due to exceptional circumstances, arrangements must be made with the teaching team (generally the lead staff member in the corresponding applied/lab class) on/before Week 9.

- ❖ Team Preparation Period (before deadline): Divide the workload among team members and complete all the tasks. For each team, you must submit the required files (e.g., report, source code, slides, and corresponding documents) before the deadline.
- ❖ Team Presentation Period (max 7 mins): For each team, you are required to present and demonstrate your design, code, results and observations during the session. All team members must be present in the applied session.
- ❖ Q&A Period (2 min): Each team member will be asked a few questions based on the submitted and/or presented work.

Note: If the required submissions are missing or any member(s) of the student team are absent, no marks will be given to the assessment.

Task 1 [Image Rotation with GPU]

In day to day photo processing, image rotation is an extremely popular/common feature allowing users to rotate (fully/parts of) their image.

In terms of mathematics in linear algebra, a rotation matrix can be used to perform the required rotation. To perform a counterclockwise rotation of θ degree (in radian) on a point with standard coordinates $v=(x,y)$ in a 2 dimensional Cartesian coordinate system, the rotation matrix R can be used to calculate the new coordinates:

$$R\mathbf{v} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} + y \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

You worked in an image (photo) processing company processing massive amount of high resolution images per day, and your team lead is interested to develop a routine to perform image rotation on high resolution large images with GPU, based on the CUDA programming model. You are being tasked to design the routine, and give a short presentation on the GPU architecture.

You are required to:

- a) Explain how image data can be transferred between the memory (e.g., DDR5 SDRAM) of a machine and the memory (e.g., GDDR5) of a GPU;
- b) Explain how the CUDA programming model operates in terms of:
 - Host (CPU) and Device (GPU) responsibilities,
 - Kernel launches with the <<<grid, block>>> execution configuration, and
 - Thread/block hierarchy (threads, blocks, grids).

Additionally, explain also the expected speed-up by modern GPU architectures.

- c) Research and explain GPUDirect Storage and why it may/may not be beneficial to image rotation.

Reminder: You are reminded to draw & include all relevant diagrams to assist your explanations.

Note: You are encouraged to develop a sample prototype in CUDA, or reference state-of-the-art image processing libraries to deepen your understanding.

Task 2 [Ethical Implications of Scaling HPC for AI]

In recent years, governments, industries, and universities worldwide have invested billions of dollars in building High Performance Computing (HPC) and AI supercomputing centres with GPUs/NPUs capable of training large language models (LLMs), large action models (LAMs), and other data-intensive AI systems.

While these advancements enable significant scientific, research and industrial innovation, they also raise complex ethical, environmental, and societal issues. These issues also include fairness, energy consumption, accessibility, dual-use risks, and the concentration of computational power.

In this task, you are required to investigate and critically discuss the *ethical implications of scaling HPCs in the age of AI*, and propose *measures or frameworks to foster responsible and sustainable parallel computing usage*.

In your presentation, your discussion should include the following topics (at least):

- a) The potential environmental and energy impact of large-scale HPC and AI training.
- b) Issues of data governance, security, and fairness when AI models depend on massive compute clusters.
- c) The socio-economic and accessibility challenges created by unequal access to HPC resources.
- d) Frameworks or principles (e.g. AI ethics guidelines, Green AI initiatives, carbon-aware scheduling, equitable HPC policies) that could promote responsible HPC use.

You should use and reference credible online resources, and cite all sources and references properly. You are also welcome to use Generative-AI tools (e.g. ChatGPT, Gemini, Copilot) to support your research, provided that you clearly declare all prompts and AI tools used.

This is an open-ended research-based question designed to assess your ability to connect technical understanding of parallel computing with its ethical and societal implications in the AI era.

Task 3 [Presentation slides and documentations]

In this task, you are required to prepare presentation slides (or documentation) to report and present your design, answers, and results from previous Tasks. You will have a maximum of 7 minutes to present your slides and approximately 1 to 2 minutes for the Q&A session.

Your presentation slides (or documentation) should have the following sections and content (at minimum):

a) (Task 1) Image Rotation with GPU [ETA: 3 min]

This section should present your answers and solutions to Task 1.

b) (Task 2) Ethical Implications of Scaling HPC for AI [ETA: 4 min]

This section should present your answers and solutions to Task 2.

c) Q&A [ETA: 1 to 2 min]

The section invites questions from the audience (TA/makers). The teaching team will ask questions related to your presentation and the submitted files.

Note: Practice makes perfect. If you are still deciding what to present and searching for information/content from the slides, chances are you will not be able to present within the time limit.

Note: You are allowed to put extra slides as an appendix (after the Q&A section) for your own convenience. Make sure they are clearly labelled so that markers will not be confused or mix them up with the other required sections.

Submission Checklist

- Task 3 - Presentation slides (or documentation)
 - The documentation can be in the format of slides, docx, PDF or an equivalent format.
 - Note that we generally do not accept any URL/hyperlinks as submissions, as the content (slides/files/documents) referenced by the link are not guaranteed to be unchanged after the deadline. To maintain fairness, markers may not mark contents solely referenced by URL/hyperlinks.
 - Please aim to keep the slides or documentation to within a 7-minute presentation.
 - Focus on emphasising the design, results and observations that would demonstrate a good understanding of the lab tasks.
- Task 1 and 2 – All associated detailed design documents, graphs, answers and results
- A separate AI declaration (in PDF format), **if not** already declared in the submitted documentation as aforementioned.

INSTRUCTIONS FOR ALTERNATIVE ASSESSMENT

This section only applies if you or your team member(s) are **granted a special consideration extension** by the special consideration team (i.e., with a valid approval notice).

1. As a reminder, note that the regular 2 day extensions would not apply for applied/lab sessions (as they are in-class assessments with group/team works).
2. Instead of reviewing student's answers (in-class) in a presentation, peer review, and/or question and answer format during the regular in-class session, **student's answers (Task 3) will be assessed based on a recorded presentation video**. The video is required to be less than 9 minutes, and markers will be marking the assessment offline.
3. Markers will only mark the first 9 minutes of the video at regular playback speed. If students decide to compress the video by speeding up the playback speed, presentation quality could be affected and marks may be deducted according to the marking rubric.
4. Instead of having in-person Q&A sessions, students will need to answer a set of fixed questions in their recording. The questions will be released in your special consideration letter. If you have special considerations approved but the letter does not contain the questions, please contact our administrative staff: Jordan Nathanael (for Clayton campus) and Jessie Leong (for Malaysia campus).
5. Apart from the recorded video file (mp4), students are still required to submit the presentation slides (PDF/pptx), associated source codes (if any), and any AI declarations/reports (PDF) listed in the regular submission checklist via Moodle before **the approved extended due date**.
6. **A late penalty** (5% per day) will still apply if you do not submit the required materials on time (against the new extended date)! As a reminder, if you submit 7 days after the extended due date, you will still receive a 0 mark.
7. AI tools are generally not allowed during the recording sessions.
8. Before you start your presentation, you are required to show your face and your student ID in the video.